

Co-integration and Causal Relationship between Energy Commodities and Energy Stock Index: Empirical Evidence from India

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Abstract. This article's primary objective is to evaluate, within the Indian setting, the degree to which energy commodities and energy stock prices correlate and are causally related. Weekly spot price data has been sourced from the Multi Commodity Exchange and the National Stock Exchange, which spans from January 2007 to December 2021. This study employs Auto-regressive Distributive (ARDL) Bound test along with Johansen co-integration approach to understand the co-integration between the series under study. The findings support the idea that the energy stock index listed on the National Stock Exchange (NSE) and the energy commodities traded on the Multi Commodity Exchange (MCX) do not exhibit any co-integration. Furthermore, to investigate the cause-and-effect connection between the energy stock index and energy commodities, Granger Causality test has been utilised, which was introduced by Toda and Yamamoto. The results demonstrate that there is no causal relationship between MCX energy commodities and the NSE energy stock index. Thus, the nonexistence of cointegration and causality between NSE energy index and energy commodities provides portfolio diversification opportunities for the investors to hedge their risk by investing in both energy commodities and energy stock index. Policymakers need to promote deeper integration among these markets by making more regulatory and accounting changes.

Keywords: Energy commodities, ARDL bound, Causal relationship, Energy, Johansen Co-integration, Stock index and Toda and Yamamoto test

1 Introduction

Crude oil and natural gas together account for well over half of the world's primary energy consumption in 2014 (Liu et al., 2017). Since the mid-1950s, crude oil has been the most significant source of energy in the world due to its high energy density, transportability, and relative abundance. According to Herodotus and Diodorus Siculus, there were oil mines in the vicinity of Babylon more than 4,000 years ago. Ignacy Lukasiewicz discovered the method for distilling kerosene from seep oil (petroleum seeps) in the 1850s, which provided an affordable alternative to whale oil. Lukasiewicz constructed the world's first commercial oil well in Poland in 1854, in response to the rapidly increasing demand for petroleum as a fuel. Lukasiewicz is also credited with inventing the kerosene lamp in 1853 and introducing the first street lanterns to Europe in the same year (Alimi & Oyediji, 2015). Natural gas, a flammable blend of hydrocarbon gases, is widely recognised for its environmentally friendly and secure energy provision. The Chinese were cognizant of the possibilities of natural gas seepage through the earth's surface as early as approximately 500 BC. The utilisation of this method involved the application of heat to sea water, resulting in the separation of salt from the liquid, so rendering it suitable for consumption. In around 1785, the United Kingdom emerged as the pioneering nation in the commercialization of natural gas, namely derived from coal, which was employed for illuminating residential dwellings and street lamps (Etiope, 2015).

Energy market is the largest commodity market in the international market. India is the 3rd largest oil and energy consumer in the world after US and China. The Energy commodities market is now going through a variety of changes and problems that are influencing its present condition. These developments and challenges are taking place on a worldwide basis. Alterations in energy costs, which are caused by changes in certain types of energy resources, are among the recent advancements that have taken place. To provide just one example, the cost of energy dropped

by 8.2% in November 2023, with coal and oil being the primary contributors to this loss. There are also important strategic choices being made by big companies in the energy sector. One example of this is the decision made by an oil major to decrease its worldwide refining portfolio in order to concentrate on large facilities that are integrated with chemical and trading enterprise operations. In addition, a monitoring authority on energy has indicated that demand growth is 'drastically' decreasing during the first quarter of this year. In addition, there is an increasing focus on issues pertaining to environmental obligations and the possibility of changes in behaviour. For instance, a number of major fossil fuel companies are now campaigning for a phaseout, and there are demands to shift away from fossil fuels, despite the fact that there are no clear dates to phase them out 1. These insights into the global energy commodity market demonstrate that it is a dynamic terrain that is characterised by price swings, strategic adjustments by important participants, and changing environmental issues. It is imperative that stakeholders in the energy industry keep a close eye on this intricate terrain and demonstrate the ability to successfully traverse it. There is increase in India's reliance on imported crude oil by 70% for the first time in 2005 and further increased by 80% in 2016 (Li and Wang, 2019). In FY22, India produced 29.7 million metric tonnes of crude oil, compared to 30.5 million metric tonnes in FY21. India has 600 million tonnes of proven crude oil reserves in 2020. India's oil consumption increased at a CAGR of 3.54 percent between 2005 and 2022. The consumption of petroleum products increased from 194 Million tonnes in FY21 to 204 Million tonnes in FY22. During FY22, the value of petroleum oil imports was Rs. 8,99,312 Crores, while the value of product exports was Rs. 3,31,615 Crores and the value of product imports was Rs. 1,82,970 Crores. Rapid economic expansion is increasing the demand for oil in production and transportation. Consequently, India's reliance on energy imports is likely to increase. It is anticipated that as income levels rise, the demand for automobiles will increase, resulting to an increase in the demand for oil and gas. Crude oil production process is one of the major inputs in an economy (refer Fig.1).

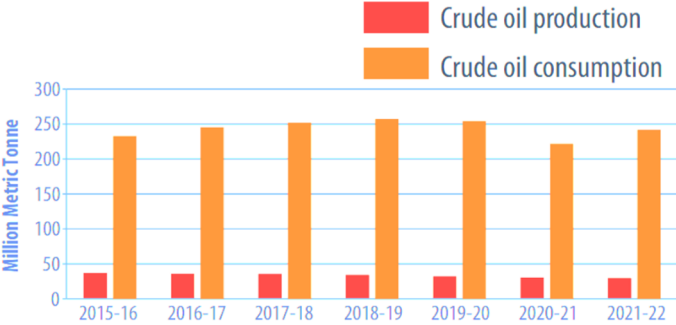


Fig 1 Crude Oil Production and Consumption (Source: www.mcxindia.com)

It appears that the world has seen a time of more crude oil price volatility based on past tendencies (Ghosh, 2011). The increased dependence of India on energy commodities includes natural gas and crude oil make it susceptible to some factors include supply interruption and variations in oil prices globally. These variances will have an impact on the equilibrium of domestic crude oil supply and demand, resulting in changes in crude oil price (Li and Wang, 2019). These differences cause the price of these commodities to become more volatile, endangering price stabilization efforts and making it more difficult for investors to predict the future price of oil (Bouri et al., 2017).

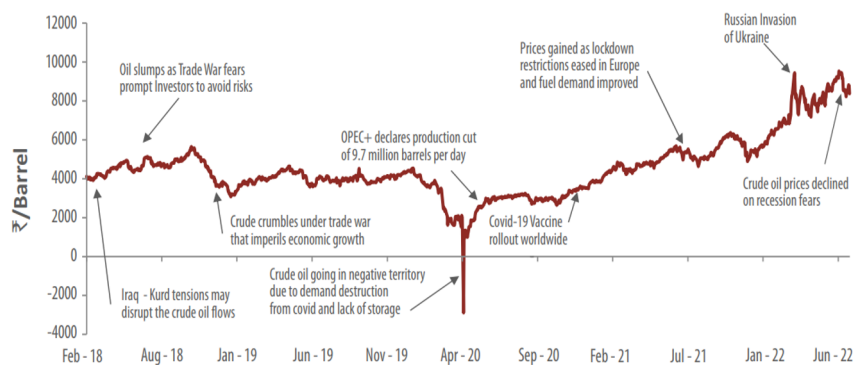


Fig.2. Crude Oil Price Movement (Source:www.mcxindia.com)

The global production of natural gas had a growth of 4.5% in the year 2021, resulting in a total production volume of 4,036.9 billion cubic metres. The United States experienced a 2% rise in natural gas production in comparison to the preceding year. In the year 2021, there was a decline of 4.9% in global natural gas consumption. However, it is noteworthy that China had a notable increase of 12.5% in its natural gas consumption, resulting in a total consumption of 378.7 billion cubic metres.

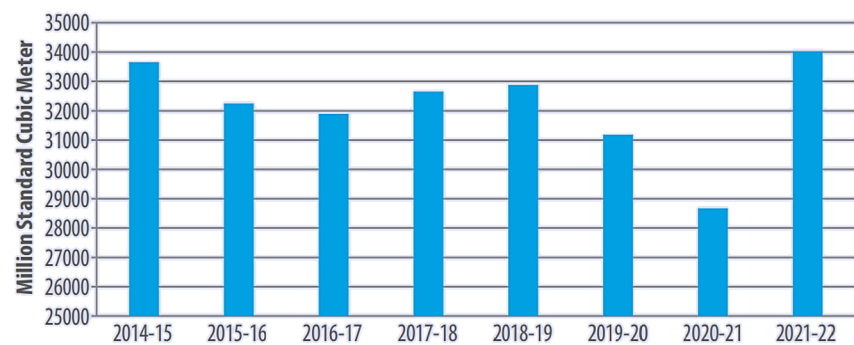


Fig 3 India's Natural Gas Gross Production (Source: www.mcxindia.com)

Figure 3 depicts that the net production of natural gas in India for the fiscal year 2021-2022 amounted to 33,131 MMSCM, representing a significant increase of 19% compared to the previous fiscal year. In contrast, the import of liquefied natural gas (LNG) reached 30,776 MMSCM, indicating a decrease of 7% compared to the previous fiscal year. The total consumption of natural gas in India during the fiscal year 2021-22 was recorded at 63,907 MMSCM, indicating a 5% increase compared to the previous fiscal year. Figure 4 illustrates the distribution of consumption across several industries throughout the fiscal year 2021-22.

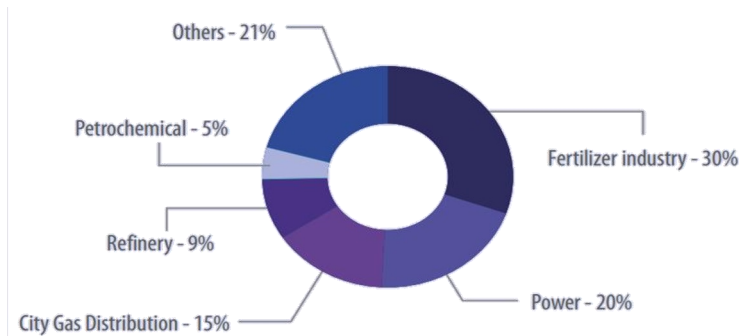


Fig 4 Industry wise consumption of Natural Gas in India (Source:www.mcxindia.com)

The escalating interconnectedness of financial markets around the world, in conjunction with the expanding financialization of commodities, has sparked significant scholarly attention towards examining the transmission of risk between prominent securities and commodity markets under certain conditions. Academic interest in studying the distribution of volatility and return correlations across different financial markets has grown substantially in the last few years (Tang and Xiong, 2010; Aboura and Chevallies, 2015; Baldi et al., 2016). During periods of market turmoil, it is imperative for portfolio managers and professionals to execute remedial measures in order to minimise the potential propagation of risk within financial markets. Empirical investigations examining the magnitude of spillovers or dynamic interrelationships among markets yield crucial insights that facilitate precise estimations of both return and volatility. The presence of correlation and volatility between energy commodities and financial markets poses significant issues, particularly for policymakers. The observed interconnectivity within these marketplaces carries substantial consequences not just for the costs of production, revenues of corporations, and rates of employment growth, but also for macroeconomic policy (Kaur and Dhiman, 2019; Yadav et al., 2023). The effect of energy commodity price swings on the return and risk characteristics of investment portfolios are, hence, something that investors must fully grasp. Vivian and Wohar (2012) have observed that a significant number of contemporary research studies are integrating raw materials as a crucial element within investing portfolios alongside other categories of equities. Traders are trying to deduce the prevailing trend and linkages between the stock and commodities markets by applying the findings of this research to the examination of price movements in these markets (Choi and Hammoudeh, 2010).

According to the efficient market hypothesis, the incorporation of news into the market has the capacity to immediately influence investor perception, resulting in the modification of their portfolios or investments. According to Tursoy and Faisal (2017), financial markets swiftly respond to these alterations. Prior studies have shown a significant lack of correlation between commodities and stock prices. Commodities are classified as an asset class since they are not closely linked to other financial assets such as bonds and equities (Elder et al., 2012). Consequently, include commodities in a portfolio has the potential to increase total investment returns. The disparity in the influence of news on commodities and equities prices is a fundamental factor that leads to a weak connection between these two types of assets. The reciprocal effect between people may be attributed to the negative influence they exert on one other (Hammoudeh & Aleisa, 2004; Reddy & Sebastin, 2009). This research aims to evaluate the correlation between commodity and stock markets by focusing on the NSE energy stock index and energy commodities. The researchers used the autoregressive distributive lag (ARDL) bound test and the Granger Causality Test, as described by Toda and Yamamoto, to determine the existence of lead-lag relationships and cointegration among the variables.

2 Literature Review

The Influence of Energy Commodities on Energy Stock Prices Energy commodities have a substantial influence on the stock prices of energy firms. These variables have a significant influence and should be taken into account by investors, policymakers, and market experts. Interconnected Relationship The correlation between energy commodities and stock prices of energy businesses is intricate and diverse. Fluctuations in the pricing of energy commodities, such as crude oil, natural gas, and coal, may significantly influence the financial performance and

value of energy firms, hence impacting their stock prices. Economic forces and factors that influence the behaviour and performance of a market. The relationship between energy commodities and energy stock prices is shaped by market dynamics, global supply and demand patterns, geopolitical occurrences, and economic circumstances. changes in crude oil prices are primarily influenced by geopolitical tensions, production levels, and global demand. These changes may have a substantial impact on the profitability and stock prices of energy businesses engaged in oil exploration, production, and refining. Inflation and Economic Trends Energy commodity prices may influence energy stock prices due to their correlation with wider economic patterns and inflation. Fluctuations in energy costs may impact the overall trajectory of inflation, since decreases in energy prices have the potential to result in a deceleration of inflation. The correlation between energy prices and inflation may impact the assessment and effectiveness of energy stocks. Hazard and Ambiguity Moreover, the correlation between energy commodities and energy stock prices is impacted by variables such as market risk preferences, general uncertainty, and risk aversion. Market players often evaluate the influence of energy commodity price swings on the financial prospects and risk characteristics of energy firms, resulting in volatility in stock prices. In conclusion To summarise, understanding the influence of energy commodities on energy stock prices is an essential field of research for comprehending the intricacies of energy markets and making informed investment choices. Stakeholders must acknowledge the interdependence between energy commodities markets and energy stock markets. They should take into account the many elements that influence this connection when analysing and making choices about energy investments. The correlation between commodities and stock prices has gained significant attention worldwide during the financial crisis of 2007-2008 (Tang and Xiong, 2010; Baldi et al., 2016). There has been extensive empirical research on the correlations between energy commodity and financial markets (Ciner, 2001; Hammoudeh and Aliesa, 2004; Magheyereh and Al-Kendari, 2007; Harri et al., 2009; Broadstock et al., 2012; Chebbi and Derbali, 2015; Kisswani and Elian, 2017; Tursoy and Fasoy, 2017). Tursoy and Faisal (2017) found that changes in oil prices have a beneficial impact on stock index values. However, this relationship is not linear, as previously noted by Ciner (2001). In order to investigate the relationships between crude oil prices and the stock markets of GCC member states, Hammoudeh and Aliesa (2004) used a Vector Autoregressive (VAR) model. A lack of direct association between these stock markets and oil prices has been discovered, indicating that these markets are unable to accurately forecast future oil prices. The financial markets of GCC member nations are impacted by the country-specific disturbances. In a study conducted by Magheyereh and Kendari (2007), The members of the Gulf Cooperation Council (GCC) did not show any consistent correlation between the equities market and crude oil prices. Nevertheless, there has been a suggestion that the equity prices of these countries are subject to a substantial nonlinear influence from fluctuations in oil prices. The incorporation of crude prices suggests a potential for predicting stock market prices. The research also used the DCC GARCH Model to look examine how changes in crude prices affect Chinese energy equities. A dynamic link between energy stocks and oil prices in China has been identified. According to Broadstock et al. (2012), there has been an observed increase in conditional correlation following the occurrence of the 2008 financial crisis. According to Soucek's (2013) findings, during times of financial crises, the link between petroleum oil and the stock market noticeably weakens. Urrutia and Malliaris (2005) present a contrasting perspective to the conclusions given by Soucek (2013), suggesting that the relationship between equities and crude oil becomes more pronounced in times of financial crises. This heightened connection diminishes the potential advantages of diversification for investors. An analysis of the correlation between the Asia-Pacific stock market and the oil market was conducted by Zhu et al. (2014). Their study found that there is only a minimal relationship between oil prices and Asian stock markets. It must be noted, nevertheless, that this link underwent a major shift after the financial crisis of 2008. There has been a significant increase in the relationship between these two markets since the financial crisis occurred. According to the study conducted by Chebbi and Derbali (2015), The phenomenon of financialization in the commodities market is a crucial factor contributing to the observed positive correlation between these two sectors. Oztec and Ocal (2016) argue that the rise in financial investors participating in the commodity market, commonly known as "financialization in commodity market," has contributed significantly to the increased correlation observed between the stock market and commodity market in recent times (De Boyrie and Pavlova, 2016). The rise in trading volume of commodity futures suggests a elevated responsiveness of the commodity market to investor sentiment in recent times. The influence of demand and supply patterns on the volatility of commodity and equities market prices has decreased.

The correlation between energy commodities and stock prices has been the subject of very few scholarly investigations in the Indian setting. Nevertheless, this subject is very significant in the academic community. The correlation between crude oil prices and stock prices was the focus of Ghosh and Kanjilal's (2016) study of the Indian market. Oil price-stock market connection was shown to be impacted by the budget deficit, according to the researchers. As the price of oil has risen, so has the cost of imported products. Subsidy payments to the government

become more substantial because of this. The outcome is that the rate of inflation inside the country has increased. The wealth effect, claims Girardi (2015), is to blame for the decline in stock market investment. In addition, Bouri et al. (2017) put out the idea of co-integrating the energy commodities market with the stock market. There is a one-way causal relationship between crude oil and the stock market. Based on the statistics, crude oil is one of India's main imports, and the country's financial markets are quite sensitive to changes in oil prices. Dutta (2017) used the implied volatility index to find a fascinating correlation between oil prices and the US energy stock market. There was a strong correlation and causal relationship between the two variables, according to the research. There is a causal link between oil and US stock performance; that is, oil affects US stock performance. This is primarily because crude oil is an essential component in the operations of US energy businesses, which are significant players in the stock market. The stock performance of these firms is similarly affected by the unpredictable fluctuations in crude oil prices.

After a thorough review of the pertinent scholarly sources, it can be inferred that the level of interconnection across these markets has experienced a notable upsurge in recent times. The commodity market has witnessed a notable surge in investor participation, leading to the recognition of commodities as financial assets that can be compared to equities and bonds. Because of commodities' increased reliance on financial markets, the interdependence of the stock and commodity markets has changed. Because commodity prices include useful information about expected economic circumstances, the correlation between commodity prices and the stock market may be used as a tool for prediction. When commodity prices rise, it may indicate that global economic demand is also on the rise, which in turn affects stock prices (Black et al., 2014). There exists a divergence of opinions among scholars about the precise conceptualization of co-integration between commodity and equities markets. The escalation of commodity prices has been widely recognised as a prominent factor contributing to elevated levels of inflation and interest rates within the economy. Consequently, this phenomenon has exerted an adverse influence on stock prices. Consequently, the establishment of a position in the commodity market can function as a risk management strategy to mitigate unforeseen price swings in the stock market (Conover et al., 2010). Hence, a lack of unanimity exists among scholars with regards to elucidating the concept of connectivity between these two marketplaces. The link between stock and commodities markets has been extensively studied in the literature. However, understanding this phenomena in the Indian context remains puzzling. Considerable study has been conducted in developed economies to investigate this subject, however, there is a noticeable lack of literature on emerging economies like India. Developing countries' financial markets vary from developed economies in terms of their volatility and speculative activity (Ping et al., 2017). The lack of empirical research is a major factor that adds to the confusion around this concept in the Indian context. Taking into account all of these features and limitations provides a theoretical framework for future inquiries. This research addresses the lack of information about the co-integration of the stock and commodity markets in India by examining energy commodities and the energy stock index.

The recent developments made by the Security and Exchange Board of India (SEBI) have increased the level of exposure of individual investors in India to commodities. India's Securities and Exchange Board of India (SEBI) has recently taken steps to integrate the country's stock and commodities markets. This involves bringing together investors, intermediaries, and the operational framework. The Bombay Stock Exchange (BSE) and the National Stock Exchange (NSE), the two main stock exchanges in India, have recently submitted applications seeking authorization to commence trading in commodities derivatives. This is a fascinating move towards the consolidation of financial markets and enhancing investor trust. SEBI has just approved the use of option contracts for commodities that are not related to agriculture. Consequently, there has been an increase in the involvement of investors in the commodities market. This study investigates the co-integration between the stock and commodity markets to gain a better understanding of how effective commodities are as a hedge against unexpected fluctuations in the stock market. It takes into account recent developments and amendments in commodity trading that aim to boost investor confidence and participation. Furthermore, the relationship between the prices of raw materials and their respective stock indices will provide valuable insights into possible replacement tactics between commodities and equities (Creti et al., 2013). As a result, they will have a greater ability to predict and be ready for unforeseen changes in the prices of commodities and stocks.

3 Research Methodology

The current investigation used a weekly monitoring of crude oil, natural gas, and the NSE energy index from January 2007 to December 2021. The data on energy commodities and the energy stock index is obtained from the official websites of MCX and NSE, respectively.

3.1 Unit Root Testing

A unit root test is a statistical method used to ascertain the presence of a unit root in a time series variable. A unit root in a time series indicates that it is non-stationary, meaning that it has a noticeable trend and lacks consistent statistical properties over its lifetime. In contrast, a stationary time series maintains a constant average, variability, and autocorrelation structure during its entire length (Kwiatkowski et al., 1992). Several unit root tests are available, with the Augmented Dickey-Fuller (ADF) test being a commonly used option (Nkoro & Uko, 2016). The Augmented Dickey-Fuller (ADF) test is a more sophisticated iteration of the Dickey-Fuller test, often used in the domains of econometrics and finance. The objective of this test is to assess the null hypothesis that a certain time series has a unit root, as opposed to the alternative hypothesis that the series is stationary. The reference is from Dickey and Fuller's work in 1979. Below is the regression equation for this test.:

$$y_t = \alpha + \beta_t + \gamma y_{t-1} + \sum_{i=1}^n \delta_i \Delta y_{t-i} + \varepsilon_t \quad (1)$$

Here y_t is the price series of individual commodity or stock index series, γ is the coefficient on the lagged level of the variable y_{t-1} , $\sum_{i=1}^n \delta_i$ is the coefficient of the lagged difference of the variable (Δy_{t-i}) and error term, denoted as ε_t , is considered to be white noise.

Researchers frequently engage in the comparison of lag order selection through the utilisation of several criteria. The objective is to identify the lag order that consistently yields a parsimonious model while maintaining a high level of fit to the data. Recognising the importance of selecting the appropriate lag order is essential as it significantly impacts the performance and interpretation of the cointegration model. Hence, it is advisable to show prudence and do comprehensive testing before making this option. The use of automated approaches, such as the Akaike Information Criterion (AIC), provides a way to speed up the process of selecting an appropriate lag order (Ivanov and Kilian, 2005).

3.2 Co-integration Testing

The ARDL Bound test, created by Pesaran, Shin, and Smith (2001), was used to examine the co-integration between energy commodities and the energy stock index. Due to its versatility and effectiveness, this test is preferred above other approaches for examining long-term co-integration. This strategy is often used, irrespective of the integration order of the variables (Bouri et al., 2017; Dutta, 2017). However, it is essential for this approach that none of the variables be of type I(2) or higher. The equation used in the ARDL co-integration test to examine the relationship between energy commodities and the NSE energy index is shown below.

$$\Delta NSE \text{ Energy} = \omega_1 + \sum_{i=1}^n \alpha_{1i} \Delta NSE \text{ Energy}_{t-i} + \sum_{i=1}^n \beta_{1i} \Delta \text{Energy Com}_{t-i} + a_1 NSE \text{ Energy}_{t-1} + b_1 \text{Energy Com}_{t-1} + \varepsilon_{1t} \quad (2)$$

$$\Delta \text{Energy Com} = \omega_2 + \sum_{i=1}^n \alpha_{2i} \Delta NSE \text{ Energy}_{t-i} + \sum_{i=1}^n \beta_{2i} \Delta \text{Energy Com}_{t-i} + a_2 NSE \text{ Energy}_{t-1} + b_2 \text{Energy Com}_{t-1} + \varepsilon_{2t} \quad (3)$$

Where Δ denotes first differenced operator, Energy Com indicates energy commodities, ε_{1t} and ε_{2t} are error terms which are white noise.

The Auto-regressive Distributive Lag test is dependent on non-standard distributed Wald tests or F statistics. This test is used to assess the co-integration between two or more series over the long run. The estimated F-statistic value using this model is compared to the provided upper and lower bound values. If the estimated value exceeds the upper bound value, the absence of cointegration is rejected as a null hypothesis. The result is inconclusive if the

estimated value fell between the upper and lower bounds. In such a scenario, the term error correction characterises the long-run co-integration of the variables. The negative and statistically significant error correction term indicates that the series have a long-run association. If the calculated F-statistic falls below the lower limit value, the absence of co-integration between the series is indicated by the null hypothesis.

3.3 Toda and Yamamoto Testing

The Toda-Yamamoto (1995) causality model is a statistical technique employed for the purpose of examining causal relationships among time series data. The Granger causality test is expanded by incorporating the potential existence of non-linear causal associations. This measure demonstrates a higher level of effectiveness compared to other standard approaches in determining the lead-lag connection. First, this approach's accuracy is not contingent on the order of integration. This test is applicable regardless of the order of integration. Second, examining co-integration prior to employing the causality model is unnecessary. This method has reduced the bias associated with unit root testing and the co-integrating properties of variables (Siami-Namini, 2017).

The Toda-Yamamoto causality model can be expressed in the following general form:

$$y_t = \alpha + \sum (\beta_i * y_{t-i}) + \sum (\Gamma_j * x_{t-j}) + \varepsilon_t$$

Y_t represents the time series of the dependent variable at time t
 X_t represents the time series of the independent variable at a certain time point, denoted as t .
 α represents the intercept term in the context of the given equation.
 β_i denotes the coefficients associated with the lagged values of the dependent variable.
 Γ_j denotes the coefficients associated with the lagged values of the independent variable. X .
 ε_t represents the error term at time t .

4 Results and Discussion

The ADF test is utilised to determine whether the series under consideration contains a unit root. The examination is conducted using both the level and first difference methods. The findings presented in Table 1 indicate the presence of unit root at the level for all the series pertaining to natural gas, petroleum oil, and the NSE energy index. Following this, each series is differenced and subsequently retested to ensure stationarity. The findings suggest that the series attains stationarity at the first difference. The ARDL Bound test operates under the assumption that every series can be integrated at order zero or one. The series should not exceed $I(2)$.

Table 1 Unit Root test Results for Energy Commodities and Energy Index

Variables	Level		First Difference	
	<i>t</i> -Statistics	Probability	<i>t</i> -Statistics	Probability
Crude Oil	−3.238846	0.2901	−22.26091	0.0000
Natural gas	−1.880784	0.1232	−24.46685	0.0000
NSE Energy	−2.428424	0.7572	−23.59602	0.0000

The Authors

Additionally, the AIC criteria implemented within the VAR system were utilised to ascertain the most advantageous lag length for the pairings of energy commodities and energy stock index. The optimal latency length for the crude oil-NSE energy and natural gas-NSE energy couples, as indicated in Table 2, is 1.

Table 2 Lag Length Selection Energy Commodities and Energy Index Pairs

Pairs of Variables	Lag Length
Crude Oil–NSE Energy	1
Natural Gas–NSE Energy	1

The Authors

ARDL Bound Test is employed to empirically analyze the long-run co-integration between energy commodities and Energy stock index and results shown in Table 3 depicts the nonexistent of co-integration between energy commodities and energy index, because F-statistics is less than the lower bound critical value. As it is evident that there is no co-integration between energy commodities and the NSE energy index, it is suggested that investors can use crude oil, natural gas, and NSE energy index as diversification tools in portfolio allocation.

Table 3. Results of ARDL Bound Test for Energy Index and Energy Commodities

Variables	F-statistics	Lower Bound	Upper Bound	Serial Correlation	Ramsey
					RESET Test
Crude Oil–NSE Energy	1.31	3.62	4.16	0.011 (0.9159)	0.254 (0.6138)
Natural Gas–NSE Energy	2.52	3.62	4.16	0.255 (0.6131)	0.021 (0.9826)

The Authors

Table 3 presents the results of the ARDL Bound diagnostic examination. The obtained outcome provides confirmation that the residuals do not exhibit serial correlation. Additionally, the Ramsey RESET test verifies the stability of the ARDL model. Stability and dependability of the ARDL model are validated by these diagnostic tests.

Table 4 Test Results of Johansen Cointegration

Variables	No. of CEs	Trace Statistics		Maximum Eigenvalue	
		Trace Stat	p-Value	Eigen Stat	p-Value
Crude oil–NSE	None	7.961393	0.4695	6.704051	0.5247
Energy	At most 1	1.257341	0.2622	1.257341	0.2622
Natural Gas–NSE	None	10.45282	0.2475	8.300667	0.3489
Energy	At most 1	2.152156	0.1424	2.152156	0.1424

The Authors

Moreover, to check the robustness of the ARDL Bound Test, Johansen Co-integration Test (Johansen, 1988) is applied. This method is comparatively convenient for researchers than other models as it allows more than one co-integrating equation between the variables. This test can be employed to the variables with the same integration order (Tursoy & Faisal, 2017). Johansen Co-integration test results are shown in Table 4. These results reconfirm the findings of the ARDL bound test. It means crude oil -NSE energy index, as well as natural gas -NSE energy index pairs, do not follow common trends in the long run. Ghosh and Kanjilal (2016) draw similar findings by examining the co-integration amongst the stock and crude oil prices by using Rank Co-integration Method. The results are similar to the studies suggesting the absence of co-integration between energy commodities and the stock market (Hammoudeh and Aliesa, 2004; Maghyereh and Kandhari, 2007 and Apergis and Miller, 2009).

The results indicate that, in the Indian context, energy commodities do not offer substantial insights that can be utilised to forecast stock prices over an extended period of time. A sustained reversal in the trajectory of two markets occurs when stochastic distress increases in one of the markets. The lack of a direct correlation between energy commodities and the stock market implies that energy commodities do not possess the capacity to forecast returns on the stock market. It appears that independent markets, such as these, do not exhibit synchronised movements. Therefore, in the absence of co-integration, the series exhibit unpredictable movements away from each other. The absence of co-integration between two financial markets implies that investors may enhance the diversification of their portfolio by include both energy commodities and the energy stock index. The research done

by Narsimuhulu et al. (2016) establishes that the Efficient Market Hypothesis states that stock prices reflect all publicly accessible information. Consequently, it is not feasible to estimate these values using the aforementioned information or the price indices of other financial markets. When two markets are not co-integrated, it indicates a breach of the efficient market hypothesis, suggesting that one market has sufficient information to predict the behaviour of other financial markets.

It is important to clarify that the limited connection between commodities and stocks stems from the different factors that affect the value of both assets. The absence of correlation between equities and commodities holds significance in the assessment of risk management strategies for constructing an optimal portfolio. The use of commodities into a portfolio mitigates the potential for incurring losses while yet maintaining a satisfactory level of return.

Table 5 Toda and Yamamoto Approach of Granger Causality Test Results

Dependent Variable	Independent Variable	Chi-square	Df	Prob.
NSE Energy index	Crude Oil	2.00E-05	1	0.9964
Crude Oil	NSE Energy index	0.469554	1	0.4932
NSE Energy index	Natural Gas	0.030634	1	0.8611
Natural Gas	NSE Energy index	0.814119	1	0.3669

The Authors

Utilising the Toda and Yamamoto method for analysing Granger causality, the causal relationship between the NSE Energy index and energy commodities (crude oil and natural gas) is ascertained. The outcomes derived from the execution of this model are presented in Table 5. The examination of the Toda and Yamamoto methodology begins by estimating the Vector Autoregressive (VAR) model, using the weekly closing price data of energy commodities and the NSE energy stock index. The Akaike information criterion, as shown in Table 2, is used to determine the lag duration. Subsequently, the augmented Vector Autoregressive (VAR) model is used to ascertain the order, with d_{max} representing the maximum integration order of the variables and p denoting the combined lag length. Given the absence of evidence to substantiate the existence of Granger causality, the null hypothesis is deemed valid in this statistical examination. The failure of the null hypothesis implies a probable existence of Granger causation between the series under study.

The study specifically concluded that there is no statistically significant relationship between the NSE energy index and energy commodities such as crude oil and natural gas. This supports the null hypothesis. This discovery implies that there is no meaningful correlation between energy commodities and the indices of the businesses they are associated with. This result is notable since previous research have shown evidence of the Granger causality connection between oil prices and the stock market (e.g., Huang et al., 1996; Sadorsky, 1999; Nwala, 2007). The results of this analysis align with other research that has questioned the existence of a cause-and-effect connection between energy commodities and the energy index (Gormus, 2012; Johnson & Soenon, 2014). Investors may mitigate portfolio risk by diversifying across both markets due to the absence of any causal feedback in either direction, suggesting that the two markets are mutually independent (Reddy and Sebastin, 2009; Yamori, 2010; Gormus, 2012).

Since there is no direct correlation between the commodities and stock markets, it is recommended to think about diversifying one's investment portfolio by containing both commodities and stocks. According to Ping et al. (2018), there has been a noticeable increase in the link between the stock market and commodities. However, it must be noted that different nations show different outcomes in this regard. The main rationale for these findings is because different countries' stock markets are fundamentally different. Differences in financial market volatility patterns and levels of speculation are two distinguishing features of stock markets in developing countries like China and India.

5 Conclusion

The recent advancements in the Indian commodity market, facilitated by the Security and Exchange Board of India (SEBI), have expanded the accessibility of commodities to private investors. The Securities and Exchange Board of

India (SEBI) has lately introduced steps to promote the financialization of the commodities market. The Securities and Exchange Board of India (SEBI) has just authorised the incorporation of option contracts in non-agricultural commodities. The introduction of the gold options contract has resulted in heightened investor participation in the commodities market, seen from the surge in daily gold turnover from 64 crores in December 2017 to 700 crore in July 2018. Currently, there are many options contracts available for soft metals, crude oil, copper, and zinc. Investors are employing these commodities as instruments for hedging against volatilities in the real economy. In November 2021, the National Stock Exchange (NSE) obtained regulatory authorization to issue futures contracts for WTI Crude Oil and Natural Gas. The introduction presents a novel investment possibility for Indian investors, aiming to enhance portfolio diversification and mitigate exposure to commodity price volatility. Additionally, it reduces dependence on international exchanges and enhances market liquidity within India, hence leading to improved pricing efficiency of essential commodities within the nation. This development holds considerable importance for the financialization of commodity market. Moreover, India exhibits a significant reliance on imports to fulfill its energy demands, with over 85 percent of its crude oil needs being sourced from external suppliers. This research looked at weekly data from 2007–2021, with an emphasis on India, to empirically assess the connection between the energy index and commodities. Utilising ARDL bound and Toda and Yamamoto tests, respectively, demonstrated the lack of long-run co-integration and the causal link between the energy index and commodities. Energy stock indices and energy commodities may defy the efficient market theory. This suggests that energy stock index pricing and the energy commodities market are both inefficient. No data transfer between marketplaces has been detected. The results might be explained by the fact that retail investors are less active in the commodities market, which could be due to a lack of knowledge and expertise. The commodities market also lacks the depth and liquidity that would be ideal because no institutional investors participate in it.

The findings of this study indicate that energy commodities had the potential to serve as instruments for achieving portfolio diversification and hedging objectives over an extended period, hence reducing the level of risk exposure. This is due to the lack of co-integration between these commodities and the energy stock index. In light of the current state of the market, it is imperative that energy stock index portfolio and asset managers pay closer attention to energy commodity historical data when making predictions using effective models like ARDL. This will allow them to effectively manage risk and anticipate market price movements. If lawmakers choose to further integrate these markets, this study may help them with further accounting and regulatory changes. Additionally, market players in the energy sector should be aware of the correlation between energy commodities prices and the energy stock index, and they should plan appropriately to lessen the impact of any negative developments.

6 References

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